



APASL STC 2026 Kumamoto

The Asian Pacific Association for the Study of the Liver

APASL Single Topic Conference 2026 Kumamoto

Liver Cancer Control

~Ending Hepatitis, Halting MASLD/MAFLD~

FIB-4 False Negatives in MASLD: Clinical Phenotypes and an Augmented Triage Model

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Disclosure of Conflict of Interest

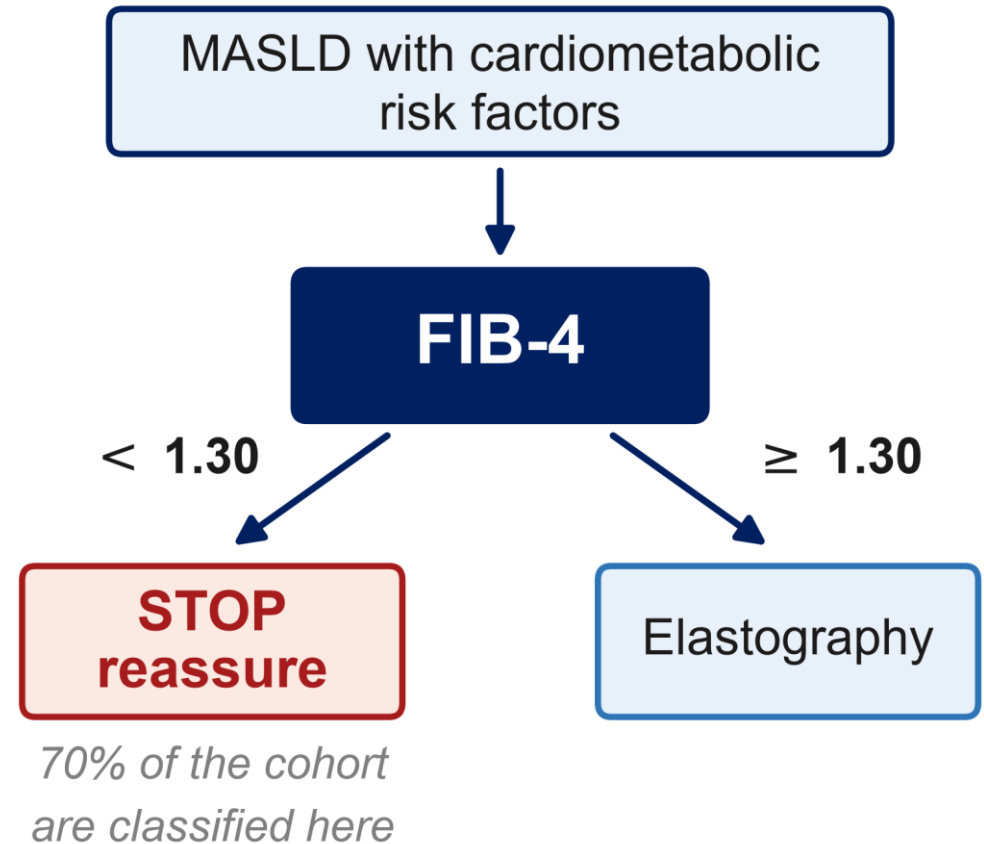
**We have nothing to declare
for this study.**

Fibrosis stage determines outcome — FIB-4 is the gate

- FIB-4 is the guideline-recommended first-line triage tool for MASLD-related fibrosis.
- It is designed to exclude advanced, F3–F4 disease.
- Below the 1.30 threshold, assessment stops and the patient is reassured.

1,390 patients — 70% of our cohort — were classified low risk and reassured.

Where does it fail, and can routine data recover it?



FIB-4 contains no measure of adiposity

FIB-4 measures

Age
AST
ALT
Platelet count



The phenotype it misses

BMI $\geq 30 \text{ kg/m}^2$
with
ALT / AST $< 40 \text{ U/L}$
transaminases look normal

The validation gap

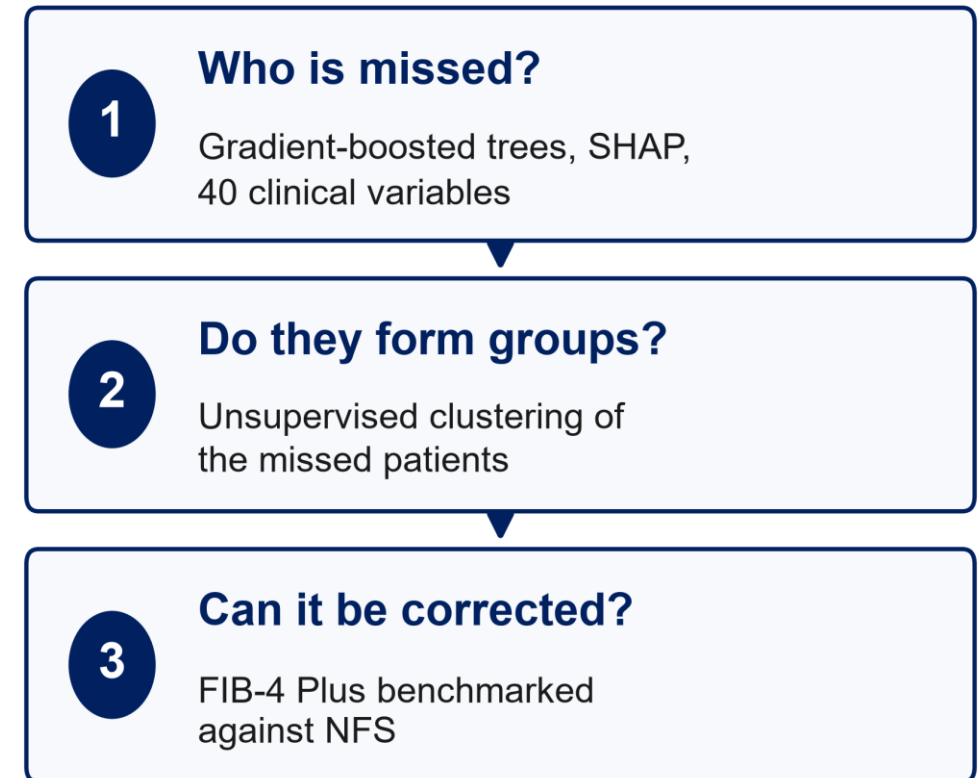
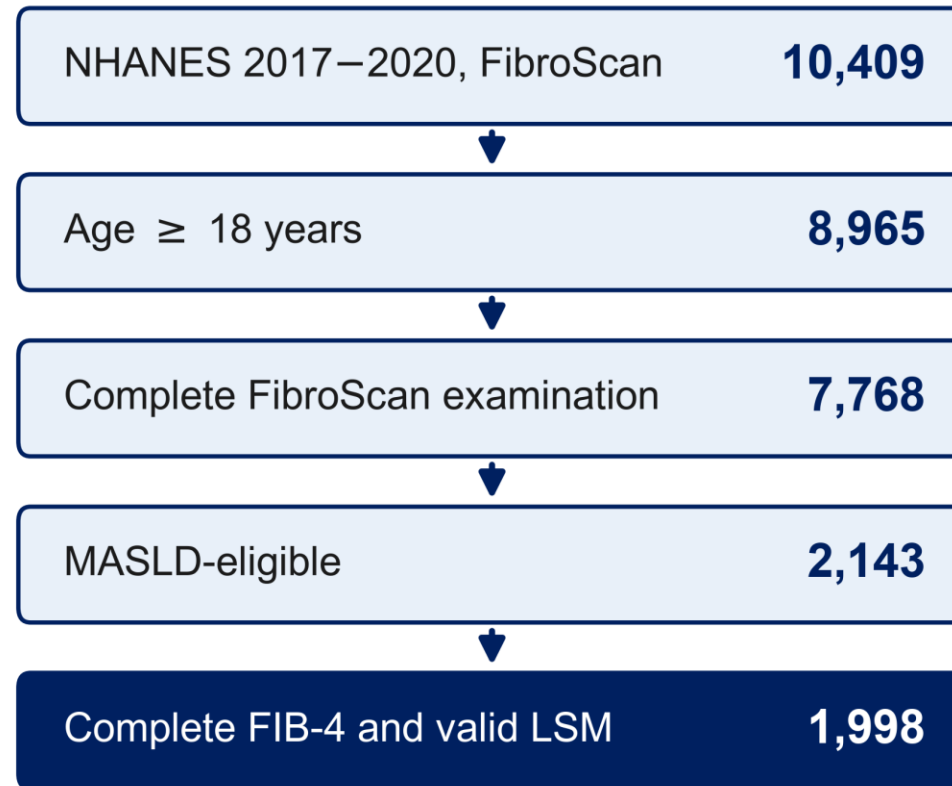
FIB-4 was validated to exclude advanced F3–F4 disease. Its performance at the F2 threshold remains poorly characterised.

Aim

Identify FIB-4 false negatives, characterise them, and develop a supplementary primary-care triage tool.

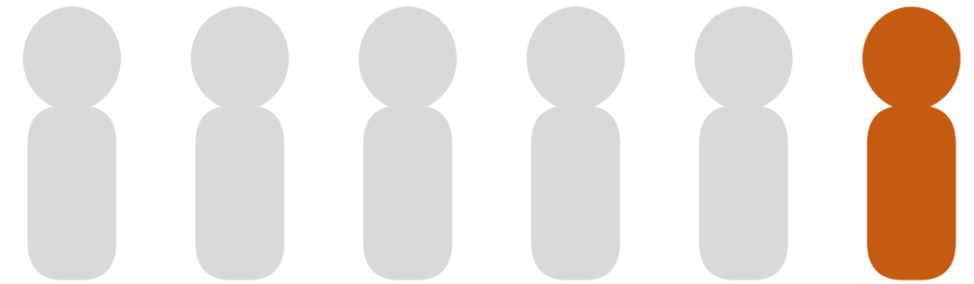
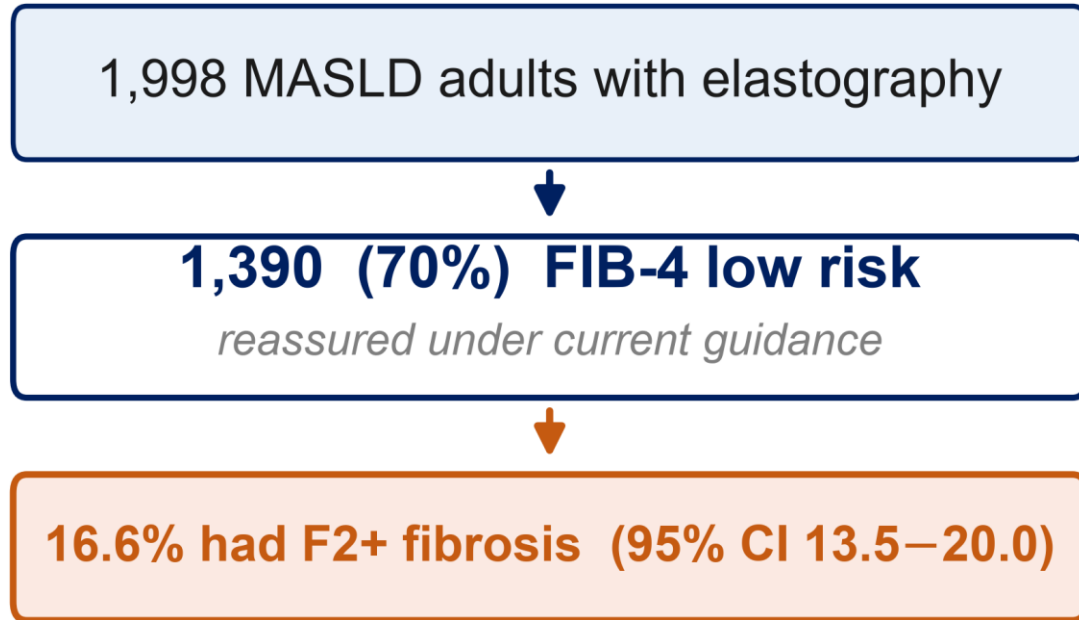
NHANES 2017–2020, with elastography on all participants

False negative = FIB-4 < 1.30 despite liver stiffness ≥ 8.0 kPa (F2+ fibrosis)



Final cohort: 1,998 MASLD-eligible adults · median age 50 years · 54% male

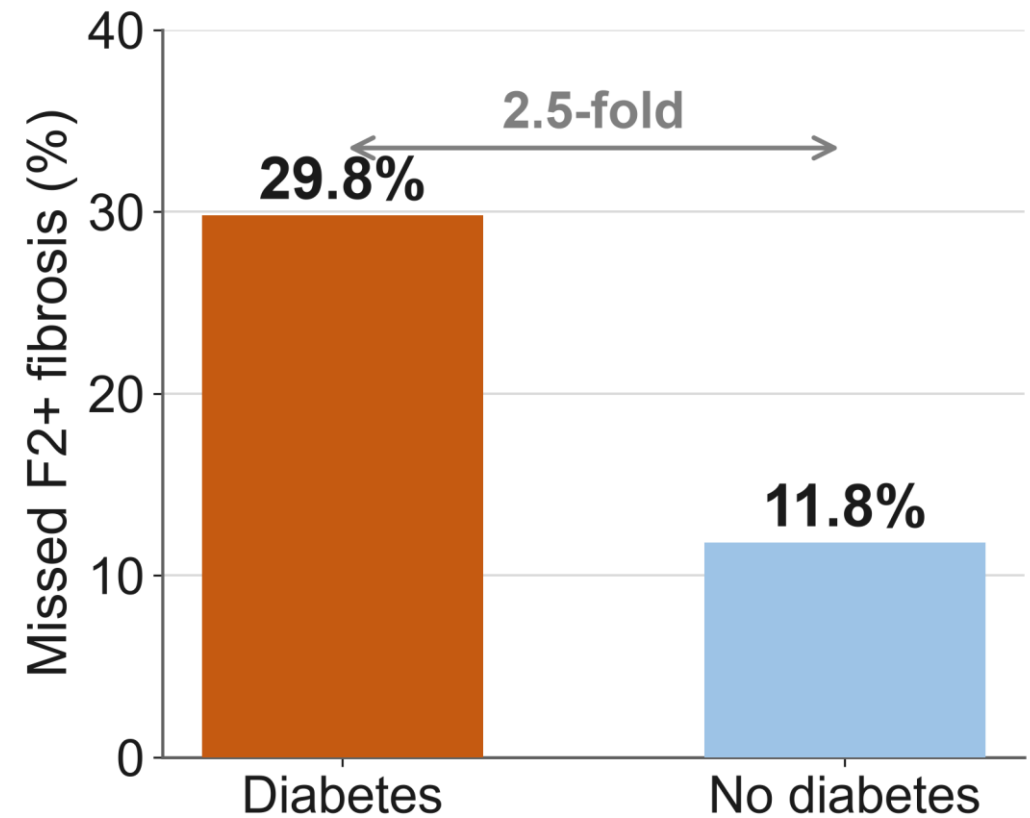
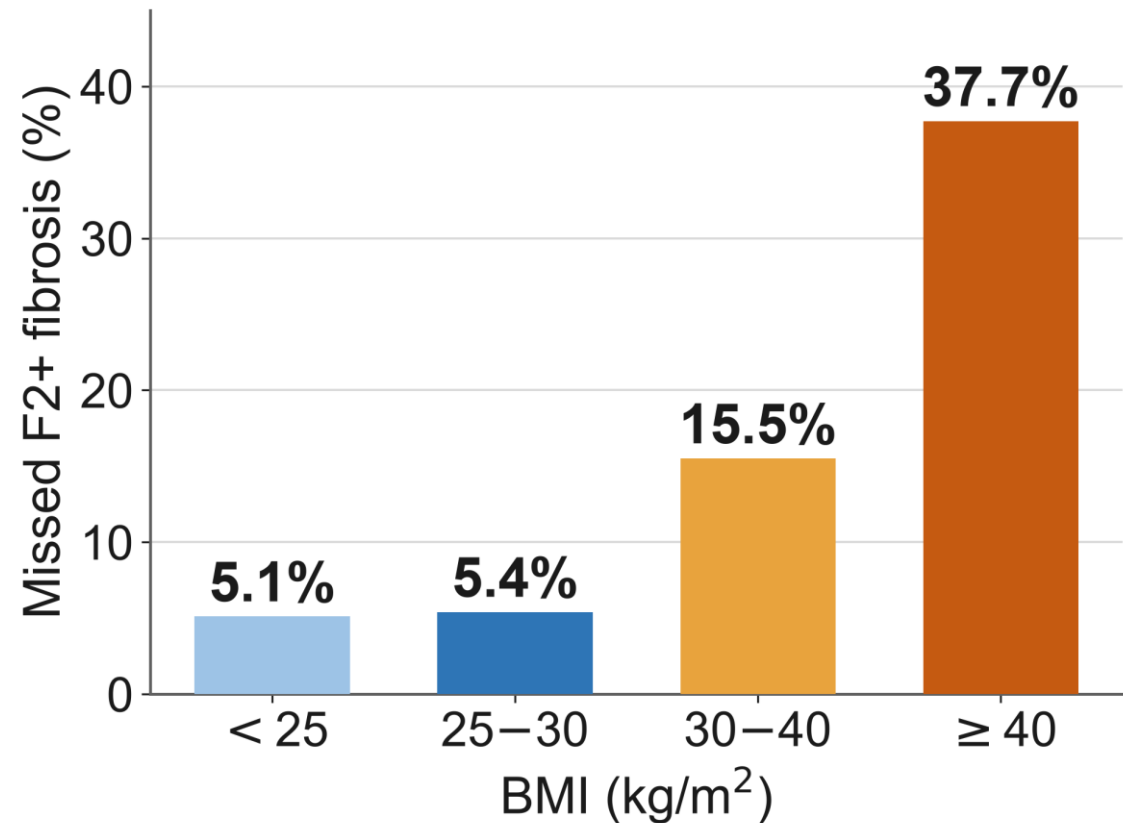
One in six low-risk patients had significant fibrosis



1 in 6 has significant fibrosis
among patients FIB-4 calls low risk

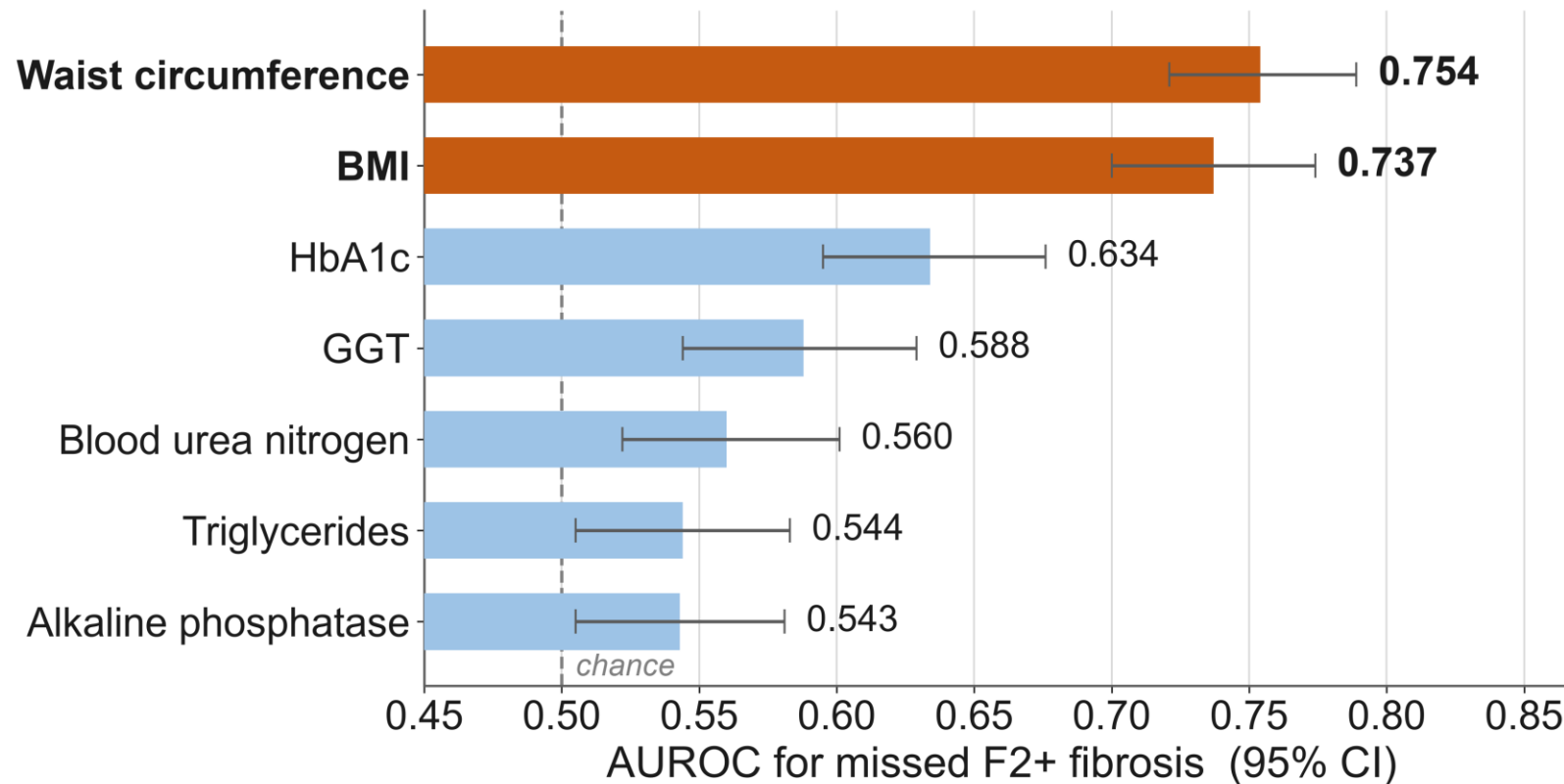
Every one of these patients was reassured by the guideline-recommended first-line test.

The miss is concentrated in obesity and diabetes



A sevenfold rise across the BMI range, and a 2.5-fold excess in diabetes — the failure is systematic, not random.

Waist circumference and BMI best predicted a missed case



Top 7 of 40 variables screened; all are absent from FIB-4.

FIB-4's own inputs are excluded — this stratum is defined by FIB-4 < 1.30, so their variance is restricted by design.

0.754

Waist circumference

0.737

Body mass index

The two leading predictors are anthropometric, absent from FIB-4, and need only a tape measure.

The missed patients form two distinct phenotypes

Severely Obese Metabolic

n = 89

BMI 46.1 kg/m²

67% female

Liver stiffness 10.5 kPa

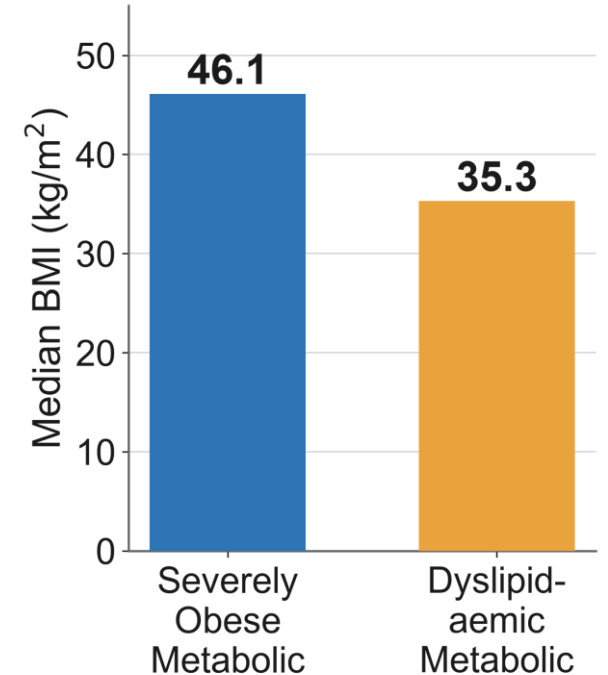
Dyslipidaemic Metabolic

n = 131

BMI 35.3 kg/m²

76% male · 44% diabetic

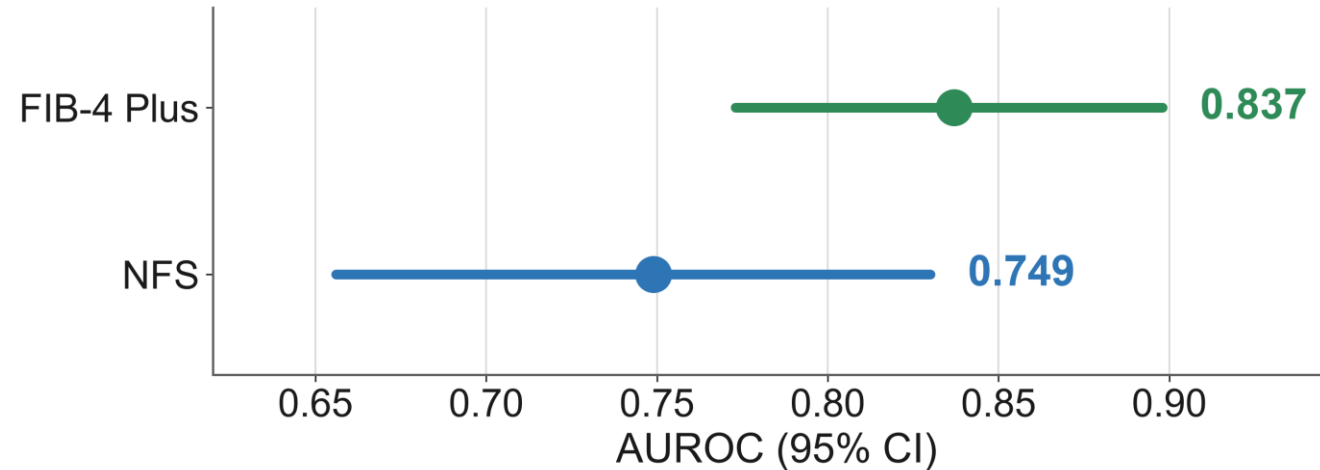
Triglycerides 168 mg/dL



Both are defined by measurements taken at every routine visit.

FIB-4 Plus: a supplementary triage step

FIB-4 Plus = FIB-4 + BMI + waist circumference + Hepatic Steatosis Index



0.837 vs 0.749

AUROC — FIB-4 Plus against NFS

p = 0.019

DeLong test

NRI 9.2%

Net reclassification improvement

In practice: every input is already recorded at a routine visit, and this triage step runs only after a low FIB-4 — the first-line test is unchanged.

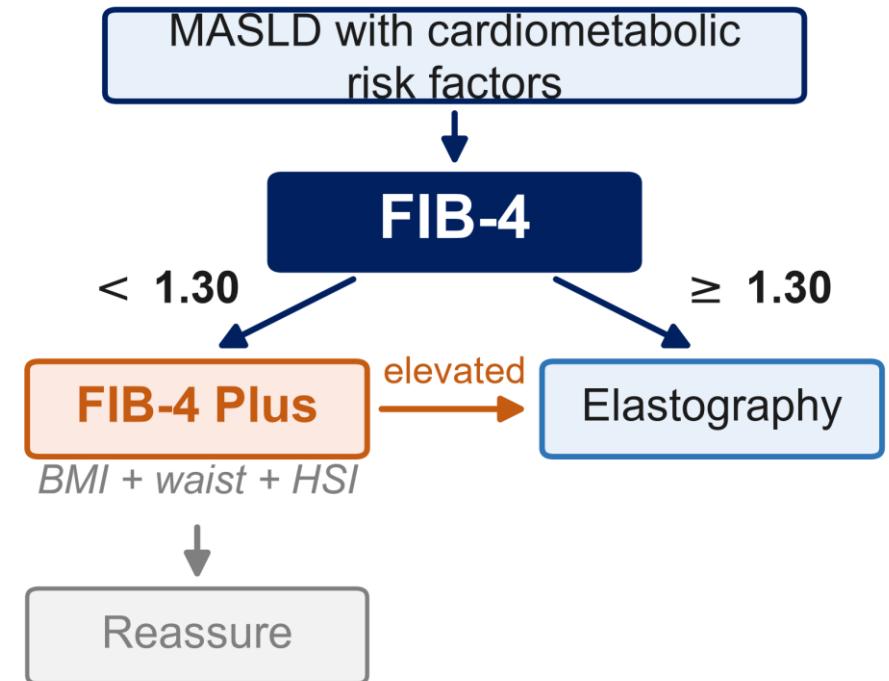
Conclusions

- FIB-4's blind spot is systematic: metabolic obesity with preserved transaminases.
- One in six low-risk MASLD patients had F2+ fibrosis, rising to 37.7% at BMI ≥ 40 .
- Supplementation with routine anthropometric and metabolic parameters recovers a substantial part of the miss.

In MASLD with BMI ≥ 30 or diabetes, a FIB-4 below 1.30 should not end the assessment.

Next step

Fibrosis occurs at lower BMI in Asian populations; validation in lean MASLD phenotypes is warranted.



One added step, no new test